Preliminary Results

Capture Theorem. Let A be a subset of $\mathbb{R}$. Any open interval containing $\sup A$ contains an element of A. Similarly, any open interval containing $\inf A$ contains an element of A.

Nested Intervals Theorem. For each $n \in \mathbb{N}$, assume we are given a closed interval $I_n = [a_n, b_n] = \{x \in \mathbb{R} : a_n \leq x \leq b_n\}$ such that

(i) $a_n \leq a_{n+1}$ and $b_{n+1} \leq b_n$ for all $n \in \mathbb{N}$, so that $I_1 \supseteq I_2 \supseteq I_3 \supseteq I_4 \supseteq \cdots$ (see the picture below).

Then $\boxed{\bigcap_{n=1}^{\infty} I_n \neq \emptyset}$.

(ii) If, in addition, $\inf\{b_n - a_n\} = 0$,

then

$$\boxed{\bigcap_{n=1}^{\infty} I_n = \{x\}, \text{ where } x = \sup\{a_n\} = \inf\{b_n\}}.$$

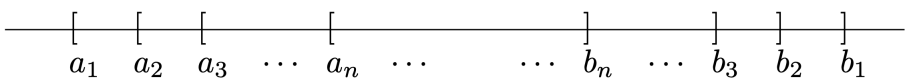


Figure 1: Intervals satisfying the hypotheses of the Nested Intervals Theorem.

Bisection Procedure.

- Begin with a closed interval $I_1 = [a_1, b_1]$.
- Bisect I_1 to obtain two closed subintervals $[a_1, \frac{a_1+b_1}{2}]$ and $[\frac{a_1+b_1}{2}, b_1]$.
- Choose one of the two subintervals above, and call it $I_2 = [a_2, b_2]$.
- Repeating this process indefinitely results in a nested sequence of intervals

  $$I_1 \supseteq I_2 \supseteq I_3 \supseteq I_4 \supseteq \cdots$$

  with $\inf\{b_n - a_n\} = 0$.
- The Nested Intervals Theorem then implies $\bigcap_{n=1}^{\infty} I_n = \{x\}$, where $x = \sup\{a_n\} = \inf\{b_n\}$.

Remark. It follows from the Capture Theorem and the nested interval structure that any open interval containing x contains I_N for some $N \in \mathbb{N}$.

To prove the Extreme Value Theorem, we first prove that a continuous function defined on a closed interval is *bounded*.

Definition. A function is said to be *bounded* if the image set $\{f(x) : x \in \text{dom} f\} \subseteq \mathbb{R}$ is bounded above and bounded below.

Example. $f(x) = \sin(x)$ is bounded, since $-1 \leq \sin(x) \leq 1$ for all $x \in \mathbb{R}$.

Lemma. If f is continuous at a, then f is bounded on an open interval containing a. (*Hint*: This follows immediately from the definition of continuity at a.)

Proof.

1. Prove the lemma above. It is an immediate consequence of the definition of continuity at a.

Theorem. If f is continuous on $[a,b]$, then f is bounded on $[a,b]$.

Proof. (By contradiction.) We use the Bisection Procedure to reach a contradiction. Suppose that f is unbounded on $[a,b]$. Then either f is unbounded on $[a,(a+b)/2]$ or f is unbounded on $[(a+b)/2,b]$ (since, otherwise, f would be bounded on their union and hence on all of $[a,b]$). Let $I_1 = [a,b] = [a_1,b_1]$ and choose $I_2 = [a_2,b_2]$ to be one of these two intervals on which f is unbounded (if f is unbounded on both, choose either one to be I_2). Now bisect I_2. Then either f is unbounded on $[a_2,(a_2+b_2)/2]$ or f is unbounded on $[(a_2+b_2)/2,b_2]$. Let $I_3 = [a_3,b_3]$ be one of these two intervals on which f is unbounded. Continuing in this way, we obtain a sequence of intervals I_n on which f is unbounded and which satisfy the hypotheses of the Nested Intervals Theorem.

2. By the Nested Intervals Theorem, $\bigcap_{n=1}^{\infty} I_n = \{x\}$, where $x = \sup\{a_n\} = \inf\{b_n\}$. Since f is continuous at x, what does the Lemma we proved above tell you?

3. As we have seen, any open interval containing x contains I_N for some $N \in \mathbb{N}$. Explain why this produces a contradiction. What do you conclude?

Theorem. (Extreme Value Theorem) A continuous function f on $[a,b]$ attains both a maximum value and a minimum value on $[a,b]$.

Proof. Since f is continuous on $[a,b]$, by the previous theorem it is bounded on $[a,b]$. We first show that f has a maximum on $[a,b]$. Since f is bounded, its image set is a subset of $\mathbb{R}$ which is bounded above, and by completeness of the real numbers it has a least upper bound. Let $M = \sup f([a,b])$. By definition of M, $f(x) \leq M$ for all $x \in [a,b]$.

4. By way of contradiction, suppose $f(x) < M$ for all $x \in [a,b]$. Then we can define a new function g by $g(x) = \frac{1}{M-f(x)}$ which is continuous on $[a,b]$ and hence bounded on $[a,b]$. Thus, there exists $K > 0$ such that

$$\frac{1}{M-f(x)} \leq K$$

for all $x \in [a,b]$. Show that this contradicts the fact that $M = \sup f([a,b])$.

5. Show in a similar way that f has a minimum on $[a,b]$.